Docker and Containers

Traditional Virtualization

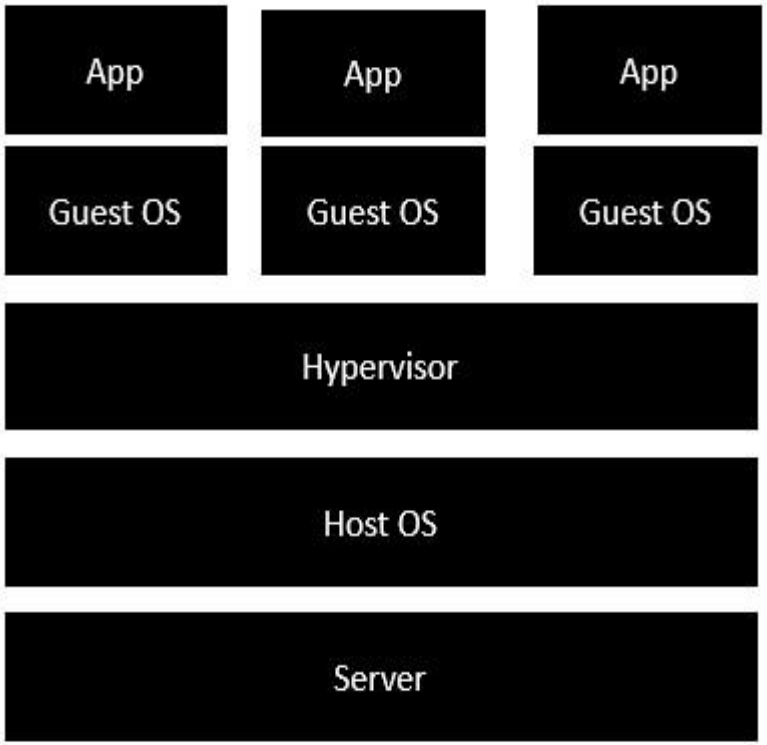
- Server is the physical server that is used to host multiple virtual machines.

- Host OS is the base machine such as Linux or Windows.

- Hypervisor is either VMWare or Windows Hyper V that is used to host virtual machines.

- <u>One would then install multiple operating systems as virtual machines on top of the existing hypervisor as Guest OS.</u>

- <u>One would then host your applications on top of each Guest OS.</u>

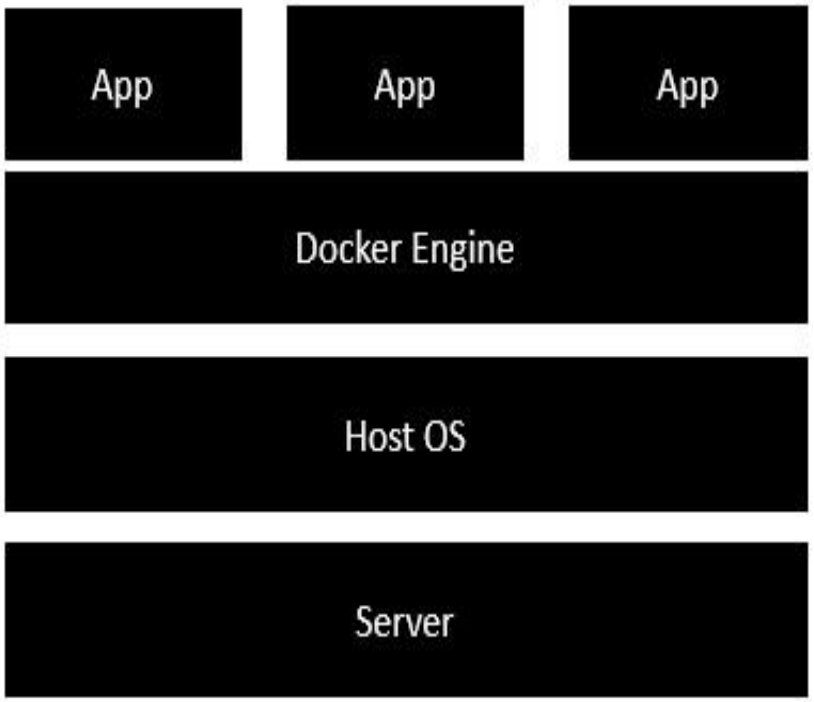


Docker – Architecture

- **Docker engine-**

is used to virtualize the guest operating system which earlier used to run in virtual machines

- All of the Apps now run as Docker containers.



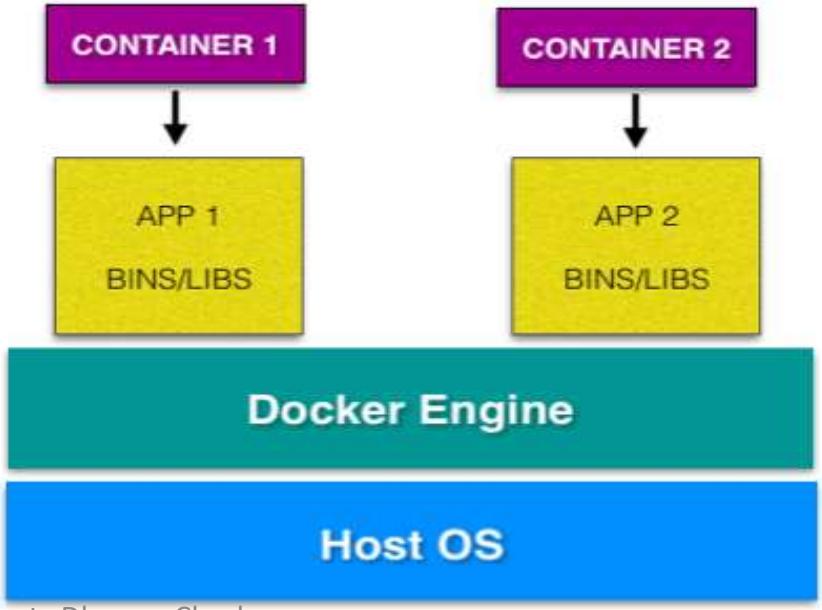
Dockers

- Dockers containers are **analogous to physical containers that you can use to store, package, and transport goods.**
- But instead of tangible goods, they're containers for software applications.

Dockers

- A container is an environment that **runs an application that is not dependent on the operating system.**

- A docker container is a portable unit of software—
- **that has the application**
- **along with the all of its associated dependency and configuration.**



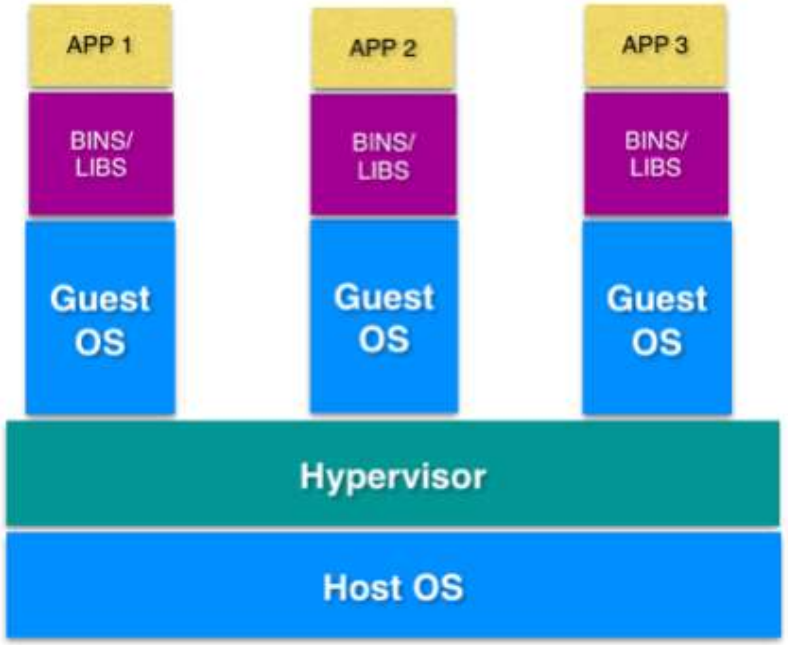
DOCKER ARCHITECTURE

What is a Container?

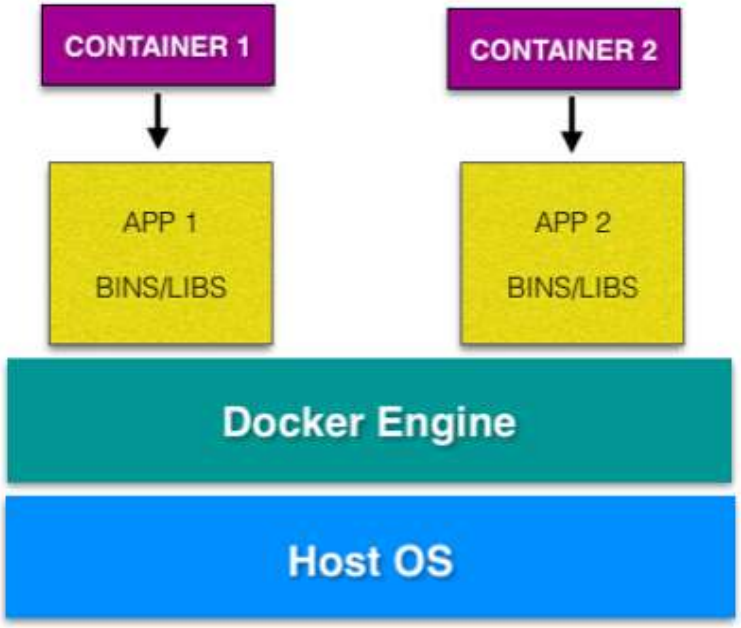
- **Each container runs isolated tasks.**
- **It cannot harm the host machine nor**
- **come in conflict with other apps running in separate containers.**

What is a Container?

- **The kernel of the host operating system**
- **serves the needs of running different functions of an app, separated into containers.**



VIRTUAL MACHINE ARCHITECTURE



DOCKER ARCHITECTURE

Virtual Machines and Containers

- **Virtual machines** run guest operating systems - the OS layer in each box.
- Resulting disk image and **application state is**
- **an entanglement of OS settings,**
- **system-installed dependencies,**
- OS security patches

Virtual Machines and Containers

- **Containers** can share a single kernel **and the only information that needs to be in a container image is the executable and its package dependencies, which never need to be installed on the host system.**
- These processes run like native processes, and can be managed individually
- Because they contain all their dependencies, there is no configuration entanglement;
- A containerized app "runs anywhere"

Learn to build and deploy your distributed applications easily to the cloud with Docker

Dockers

- *An open-source project that automates the deployment of software applications inside* ***containers*** *by providing an additional layer of abstraction and automation of* ***OS-level virtualization*** *on Linux.*

- Wikipedia

Dockers vs Containers

- Docker is a software platform that packages software into containers.
- **Docker images are read-only templates that contain instructions for creating a container**
- A container in Docker is an isolated virtual environment that runs independently. It packs all components required to run an application, including dependencies, libraries, and code.

Dockers vs Containers

- **Think of a Docker container as an apartment unit** within a building. It contains all the necessary facilities for the tenant to live independently of their neighbors, which are the runtime, libraries, and code.
- Apartment-Container
- Tenant-Application

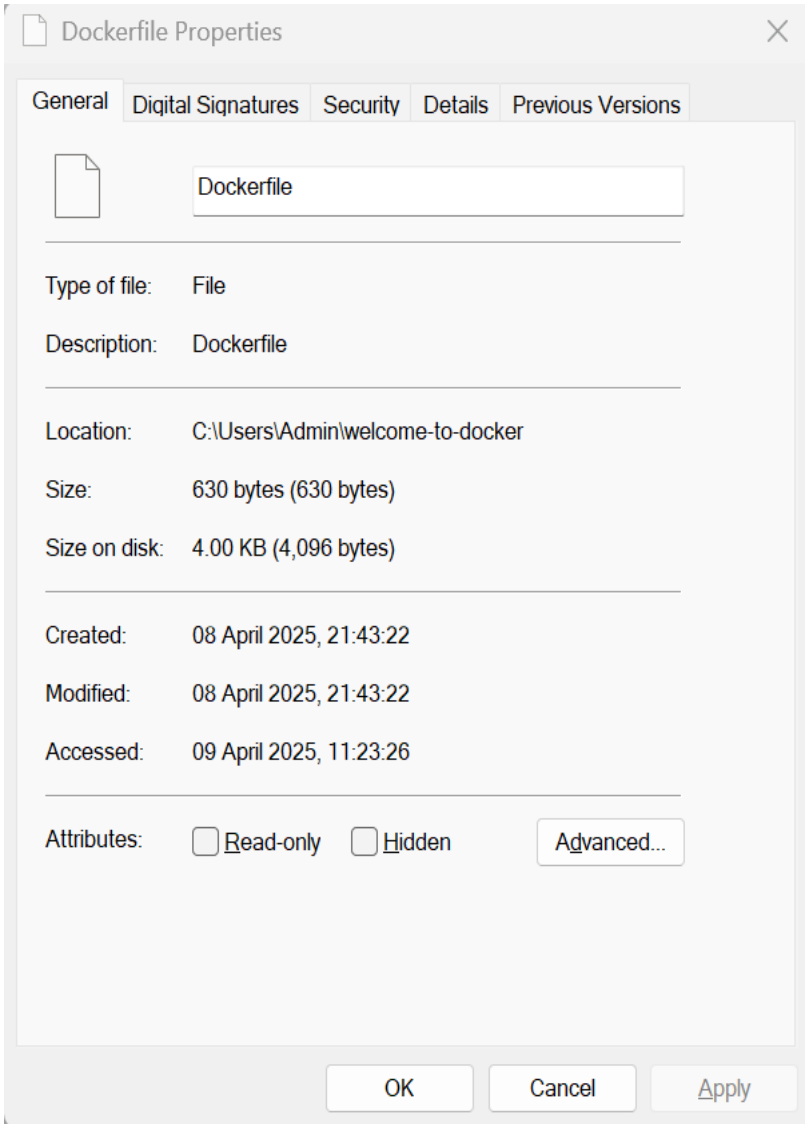
- If Docker containers are apartment units within a building, **Docker images serve as the floor plan.** It specifies the components required to build the room and the furniture needed for the tenant to live comfortably.

Docker File

- A Dockerfile is a text-based document that's used to create a container image. It provides instructions to the image builder on the commands to run, files to copy, startup command, and more.

Docker File

- A **Dockerfile** is simply a text-based file with no file extension that contains a script of instructions.
- Docker **uses** this script to **build** a **container image**.



Docker File

Code

```
# Use an official Python runtime as a base image
FROM python:3.12

# Set the working directory in the container
WORKDIR /app

# Copy the current directory contents into the container at /app
COPY . /app

# Install any needed packages specified in requirements.txt
RUN pip install --no-cache-dir -r requirements.txt

# Run the application
CMD ["python", "app.py"]
```

Docker File

- A Dockerfile typically follows these steps:
- **Determine your base image**
- **Install application dependencies**
- **Copy in any relevant source code and/or binaries**
- **Configure the final image**

Code

```
# Use an official Python runtime as a base image
FROM python:3.12

# Set the working directory in the container
WORKDIR /app

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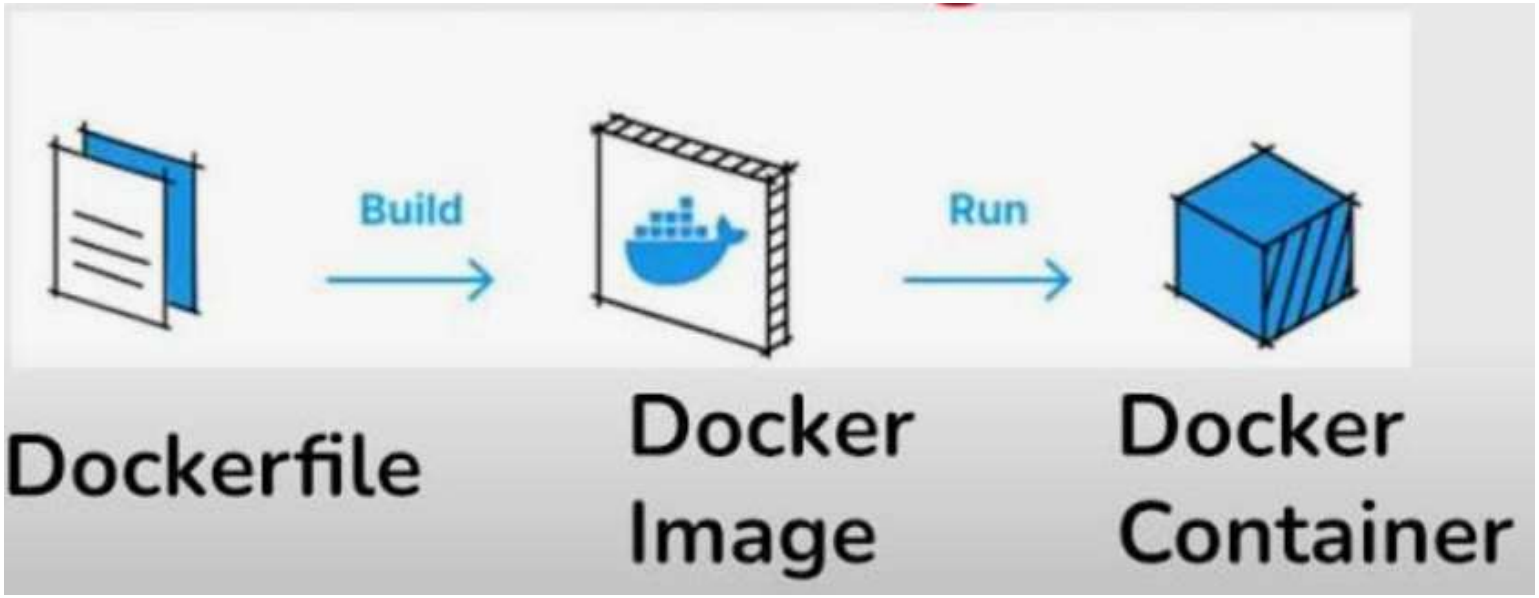
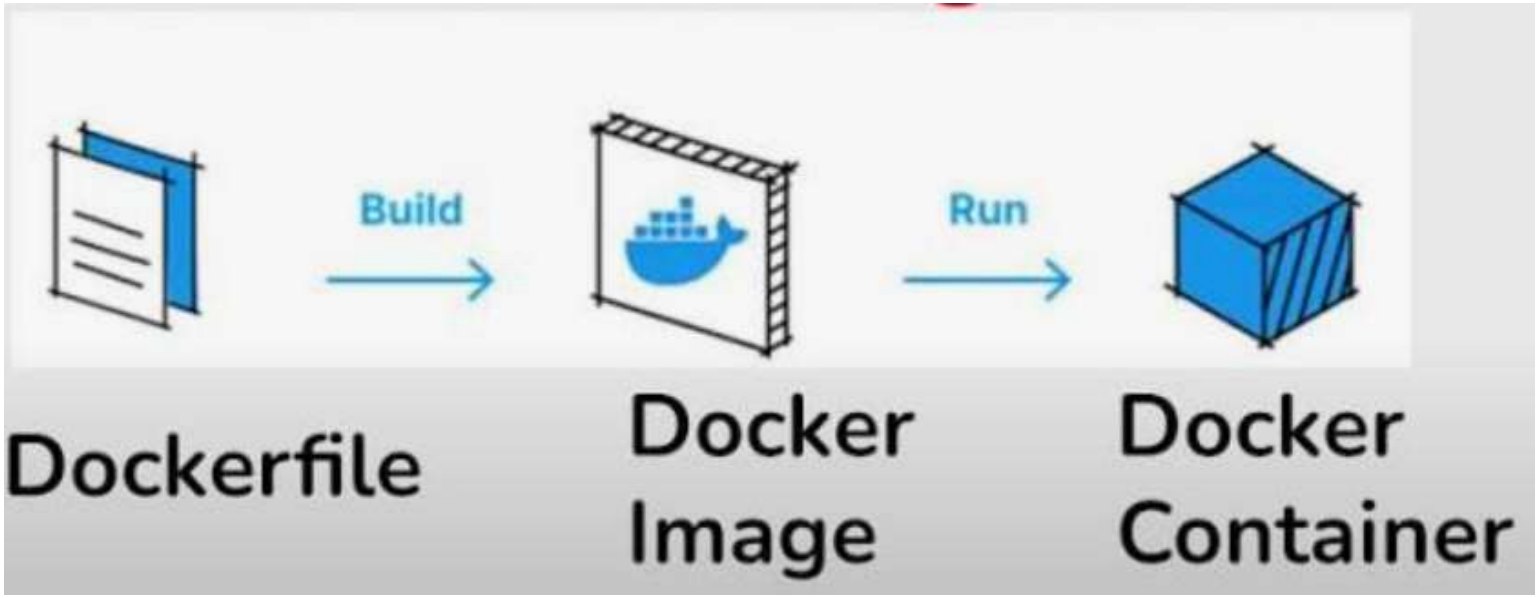
Docker File

- Using Docker File to build a Docker Image->In turn to run a Docker Container



Docker File

- Using Docker File to build a Docker Image->In turn to run a Docker Container



Containers

- Containers offer a **logical packaging mechanism in which applications can be abstracted from the environment** in which they actually run.
- This decoupling allows container-based applications to be deployed easily and consistently, regardless of whether the target environment is **a private data center, the public cloud, or even a developer's personal laptop.**
- This gives developers the ability **to create predictable environments that are isolated from the rest of the applications and can be run anywhere.**

Docker commands

- Set up Docker on Mac, Linux and Windows.

- Once you are done installing Docker,
- Test your Docker installation by running the following:

```
$ docker run hello-world

Hello from Docker.
This message shows that your installation appears to be working correctly.
...
```

Docker commands

- Lets run a Busybox container on our system and get a taste of the docker run command.
- To get started, let's run the following in our terminal:

```
$ docker pull busybox
```

- **The pull command fetches the busybox image from the Docker registry and saves it to our system.**

Docker commands

```
$ docker pull busybox
```

- *Note: Depending on how you've installed docker on your system, you might see a* permission denied *error after running the above command.*
- ***If you're on a Mac, make sure the Docker engine is running.***
- ***If you're on Linux, then prefix your*** **docker** ***commands with*** **sudo.**

- You can use the docker images command to see a list of all images on your system.

```
$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             VIRTUAL
busybox             latest              c51f86c28340        4 weeks ago         1.109 MB
```

Docker Run

- Great! Let's now run a Docker **container** based on this image. To do that we are going to use the docker run command.

```
$ docker run busybox
$
```

- Wait, nothing happened! Is that a bug? Well, no. Behind the scenes, a lot of stuff happened.

Docker Run

```
$ docker run busybox
$
```

- When you call run, the Docker client finds the image (busybox in this case), loads up the container and then runs a command in that container.
- When we executed docker run busybox, **we didn't provide a command, so the container booted up, ran an empty command and then exited.**

Docker Run

```
$ docker run busybox echo "hello from busybox"
hello from busybox
```

- Nice - finally we see some output.
- In this case, the **Docker client dutifully ran the echo command in our busybox container** and then exited it.
- If you've noticed, all of that happened pretty quickly. Imagine **booting up a virtual machine, running a command and then killing it. Now you know why they say containers are fast!**

Docker ps

- Ok, now it's time to see the docker ps command.
- The docker ps command shows you **all containers that are currently running.**

```
$ docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
```

- Since no containers are running, we see a blank line.
- Shows **only running containers**
- Lists containers that are **currently active**
- Stopped or exited containers are **not shown**
- **A list of running containers can be seen using the docker ps command.**

Docker ps -a

- **docker ps –a**

```
$ docker ps -a
CONTAINER ID      IMAGE          COMMAND          CREATED          STATUS
305297d7a235      busybox        "uptime"         11 minutes ago   Exited (0)
ff0a5c3750b9      busybox        "sh"             12 minutes ago   Exited (0)
14e5bd11d164      hello-world    "/hello"         2 minutes ago    Exited (0)
```

- Shows **all containers**
  - Running
  - Stopped
  - Exited
  - Created but never started
- **Do notice that the STATUS column shows that these containers exited a few minutes ago.**

docker ps

```
Admin@16L-18 MINGW64 ~/welcome-to-docker (main)
$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

Admin@16L-18 MINGW64 ~/welcome-to-docker (main)
$ docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS                     PORTS     NAMES
f901fef07ed0   welcome-to-docker:latest   "docker-entrypoint.s…"   43 minutes ago   Exited (0) 5 minutes ago           SDCContainer1
```

From our Implementation

Docker Run

- You're probably wondering if there is a way to run more than just one command in a container. Let's try that now:

```
$ docker run -it busybox sh
/ # ls
bin   dev   etc   home  proc  root  sys   tmp   usr   var
/ # uptime
 05:45:21 up  5:58,  0 users,  load average: 0.00, 0.01, 0.04
```

- Running the run command with the -it flags attaches us to an interactive tty in the container.
- **Now we can run as many commands in the container as we want.**

Docker Run

- let's quickly talk about deleting containers.
- We saw above that we can still see remnants of the container even after we've exited by running docker ps -a.
- You'll run docker run multiple times and leaving stray containers will eat up disk space.
- Hence, as a rule of thumb, I clean up containers once I'm done with them.

Docker rm

- To do that, you can run the docker rm command. Just copy the container IDs from above and paste them alongside the command.

```
$ docker rm 305297d7a235 ff0a5c3750b9
305297d7a235
ff0a5c3750b9
```

- On deletion, you should see the IDs echoed back to you.

Docker rm

- If you have a bunch of containers to delete in one go, copy-pasting IDs can be tedious.
- In that case, you can simply run -

```
$ docker rm $(docker ps -a -q -f status=exited)
```

- This command deletes all containers that have a status of exited.
- the -q flag, only returns the numeric IDs and
- -f filters output based on conditions provided.

Docker prune

- In later versions of Docker, the docker container prune command can be used to achieve the same effect.

```
$ docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
4a7f7eebae0f63178aff7eb0aa39f0627a203ab2df258c1a00b456cf20063
f98f9c2aa1eaf727e4ec9c0283bcaa4762fbdba7f26191f26c97f64090360

Total reclaimed space: 212 B
```

Docker Terminology

- Docker Hub - A registry of Docker images. You can think of the registry as a directory of all available Docker images. If required, one can host their own Docker registries and can use them for pulling images.

Docker Terminology

- Images - The blueprints of our application which form the basis of containers. In the demo above, we used the docker pull command to download the busybox image.
- Containers - Created from Docker images and run the actual application. We create a container using docker run which we did using the busybox image that we downloaded.
- A list of running containers can be seen using the docker ps command.

Docker Terminology

- Docker Daemon - The background service running on the host that manages building, running and distributing Docker containers. The daemon is the process that runs in the operating system which clients talk to.
- Docker Client - The command line tool that allows the user to interact with the daemon. More generally, there can be other forms of clients too - such as Kitematic which provide a GUI to the users.